A Notion of Discontinuous Feedback

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Abstract

This short note discusses a notion of discontinuous feedback which is appealing from a theoretical standpoint because, on the one hand, solutions of closed-loop equations always exist, and on the other hand, if a system can be stabilized in any manner whatsoever, then it can be also stabilized in our sense. Moreover, the implementation of this type of feedback can be physically interpreted in terms of simple switching control strategies. A recent result obtained by the authors together with Clarke and Subbotin is described, which employs techniques from control-Lyapunov functions, nonsmooth analysis, and differential game theory in order to provide a general theorem on feedback stabilization.

1 Discontinuous Feedback

It is known that, in general, in order to control nonlinear systems one *must* use switching (discontinuous) mechanisms of various types. Of course, time-optimal solutions for even linear systems often involve such discontinuities, but for linear systems most control problems usually admit also (perhaps suboptimal) continuous solutions. However, when dealing with arbitrary systems, discontinuities are unavoidable *even if no optimality objectives are imposed*. This was systematically discussed for one-dimensional systems in the early paper [20] (see also [23]). Perhaps the best-known result regarding the nonexistence of

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continuous state feedback is what has become known as the "Brockett necessary condition" (cf. [1]), which imposes severe restrictions on the right-hand side $f(x,u)$ for it to be true that the origin of the system

$$\dot{x} \;=\; f(x,u) \tag{1}$$

(in any dimension) can be stabilized under continuous feedback, that is, so that there exist any continuous k so that the origin of

$$\dot{x} \;=\; f(x,k(x)) \tag{2}$$

is asymptotically stable. An elementary exposition of this and related results can be found in Section 4.8 of the textbook [17] (see also the survey paper [16]). Faced with the theoretical impossibility of having feedback laws of the form $u = k(x)$, k a continuous function, various alternatives have been proposed. (Objectives other than state stabilization, such as tracking and disturbance rejection, output feedback, or adaptive and robust variants, also lead to the necessity of discontinuous controllers; the study of stabilization is merely the first step in the search for general controller structures.)

One way of overcoming the limitations of continuous feedback is to look for *dynamic* feedback laws, that is to say, to allow for additional (memory) variables in controllers. This was the approach suggested, and shown to be powerful enough for all one-dimensional systems, in [20], which demonstrated that a smooth *time-varying* (and, though not stated explicitely, periodic in t) controller $u = k(t,x)$ always exists. Far more interesting was the result discovered by Coron in [8], who showed that such feedback stabilizers always exist (in any dimension) provided that the original system is completely controllable and has "no drift" ($f(x,0) = 0$ for all states). (This method is closely related to the use of classical gradient descent and Newton numerical algorithms for control, see [18]).) However, for the general case of not exactly-controllable systems, and possibly with drift, no such dynamic or time-varying solutions are known, so the interest in *discontinuous* feedback laws $u = k(x)$ arises.

The first positive general theorem along those lines was due to Sussmann, who established in the early 1980s the existence of a general type of *piecewise analytic* feedback for the stabilization of controllable analytic systems; see [23]. His work, which was partly motivated by related issues arising in optimal control synthesis via the dynamic programming paradigm, provided an existence proof for feedback stabilizers for a relatively large class of systems. At around the same time, one of the coauthors of the present paper proposed a different approach, based on a general theory of controllers built out of finite automata and linear systems, in [14], using *constant-rate sampling* to deal with rather arbitrary continuous nonlinear systems. That work took into account constraints due to partial state measurements, and proposed a synthesis using what would be today called "hybrid" controllers; an expository survey was recently prepared for the proceedings of a workshop, see [19] and references given there. However, this approach was not based on true continuous-time feedback, but rather, as remarked, was basically a discrete-time theory. A general theory was lacking.

In this note, we outline the new formulation of discontinuous feedback provided in the recent paper [3], stating the relevant definitions and main results, and discuss the implications for control of various types of systems.

2 Theoretical Difficulty: Defining "Solution"

The study of discontinuous feedback leads to a major theoretical difficulty, namely the need to *define the meaning of solution* $x(\cdot)$ of the differential equation (2). As a trivial illustration, consider the one-dimensional equation $\dot{x} = -\operatorname{sign}(x)$, where we write $\operatorname{sign} x = 1$ if $x \geq 0$ and $\operatorname{sign} x = -1$ if $x < 0$ (this system could have arisen, for example, from using an on/off controller $u(t) \in \{0, 1\}$ for the system $\dot{x} = 1 - 2u$). It is easy to see that there is no possible continuous function $x(t)$ solving this equation when starting at $x(0) = 0$ (in other words, so that $x(t) = -\int_0^t \operatorname{sign} x(s) ds$ for small $t>0$). Thus no "solution" in the classical sense is possible for this example, and in general for discontinuous equations. The most often proposed definition of generalized solution of (2) is due to Filippov, who introduced in [11] an approach based on the solution of a certain associated differential inclusion. It is known, however, that *in general it is impossible to find feedback stabilizers that stabilize in the sense of Filippov*. This negative fact follows from the results in [13, 9], which amount to the statement that the existence of a discontinuous stabilizing feedback in this sense again implies the Brockett necessary condition cited above.

In [3], the following alternative is proposed. It is based on the concept of discontinuous feedback control for positional differential games introduced by Krasovskii and Subbotin in [12]. We want to define the trajectories that arise when using a feedback $k(x)$, that is, a function $k : \mathbb{R}^n \to \mathbb{U}$ from the state-space to the control-value set.

We do so by first considering, for any given partition $\pi = \{t_i\}_{i \geq 0}$ of $[0, +\infty)$, that is, for any infinite sequence consisting of numbers $0 = t_0 < t_1 < t_2 < \ldots$ with $\lim_{i\to\infty} t_i = \infty$, the *$\pi$-trajectory* of (2) *starting from* x_0, obtained as follows: on each interval $[t_i, t_{i+1}]$, the initial state is measured, $u_i = k(x(t_i))$ is computed, and then the constant control $u \equiv u_i$ is applied until time t_{i+1}, when a new measurement is taken. In other words, we use sampled control with switching instants $0 = t_0 < t_1 < t_2 < \ldots$. More precisely, for each i, $i = 0, 1, 2, \ldots$, one solves recursively

$$\dot{x}(t) \;=\; f(x(t), k(x(t_i)))\,,\; t \in [t_i, t_{i+1}] \tag{3}$$

using as initial value $x(t_i)$ the endpoint of the solution on the preceding interval, and starting with $x(t_0) = x_0$. The ensuing π-trajectory may fail to be defined on all of $[0, +\infty)$, because of possible finite escape times in one of the intervals, in which case we only have a trajectory defined on some maximal interval, but it is always defined, as in the classical case, for at least small enough intervals.

Technically, we assume in this definition that the control-value set $\mathbb{U}$ is a locally compact metric space (for instance, $\mathbb{U} = \mathbb{R}^m$ for some m) and that the mapping $f : \mathbb{R}^n \times \mathbb{U} \to \mathbb{R}^n : (x, u) \mapsto f(x, u)$ is continuous, and locally

Lipschitz on x uniformly on compact subsets of $\mathbb{R}^n \times \mathbb{U}$. Moreover, when defining feedback, we assume that the function $k : \mathbb{R}^n \to \mathbb{U}$ is locally bounded, meaning that its values on each bounded set are in some a compact subset of $\mathbb{U}$ (there are no "infinite singularities"). For each partition, we consider its *diameter* $d(\pi) := \sup_{i\geq 0}(t_{i+1} - t_i)$.

With these assumptions, and provided the trajectories stay bounded (which our theorems do insure) it is easy to prove that, for any sequence of trajectories corresponding to partitions π_k with diameters converging to zero, there is a convergent subsequence. We think of the limit of any such sequence as a "generalized solution" of the closed-loop equation (2). *Thus solutions always exist in this sense* (though, as with continuous but not necessarily Lipschitz feedback, they may fail to be unique). For ease of reference, we call here such generalized solutions "s-solutions" (for "sampling" or "switching"). It is not difficult to see that if k is continuous, every s-trajectory is a solution in the usual sense, so this indeed generalizes the usual notion.

It is important to emphasize the interpretation of s-solutions: they represent the limits of trajectories arising from high-frequency sampling when using the feedback law $u = k(x)$. This type of interpretation is somewhat analogous, at least in spirit, to the way in which "relaxed" controls are interpreted in optimal trajectory calculations, namely by high-frequency switching of approximating regular controls.

Now that the definition of solution has been given, we can turn to stating the main result of [3]. We present this result in somewhat different but equivalent terms to the formulation in that paper (in the paper, π-trajectories are used instead of the limiting s-solutions). The objective in giving the result, on stabilization, is to show that the notion is useful in that it permits solving an interesting problem. Research in progress deals with the development of extensions to other control problems.

3 A General Result on Stabilization

We define next what we mean by a feedback stabilizer. Intuitively, this is a feedback law which, for fast enough sampling, drives all states asymptotically to the origin and with small overshoot. (The more precise statements in [3] provide very precise estimates on maximal inter-sampling times, overshoot vs accuracy, etc.)

We say that the feedback $k : \mathbb{R}^n \to \mathbb{U}$ *s-stabilizes* the system (1) if for each pair $0 < r < R$, there exist $M = M(R) > 0$ and $T = T(r, R) > 0$ such that, for any initial state x_0 such that $|x_0| \leq R$, every s-solution $x(\cdot)$ of (2) starting from x_0 is defined for all $t \geq 0$ and it holds that:

1. $|x(t)| \leq r$ for all $t \geq T$;
2. $|x(t)| \leq M(R)$ for all $t \geq 0$;
3. $\lim_{R\downarrow 0} M(R) = 0$.

Observe that these three properties represent, respectively, uniform attractiveness, overshoot boundedness, and Lyapunov stability of the closed-loop system.

We also need to recall the definition of global, null-asymptotic controllability.

The system (1) is *asymptotically controllable* if (attractiveness): for each $x_0 \in \mathbb{R}^n$ there exists some control u (a locally essentially bounded measurable function $u : [0, +\infty) \to \mathbb{U}$) such that the trajectory $x(t) = x(t; x_0, u)$ (the solution of (1) at time $t \geq 0$, with initial condition x_0 and control u) is defined for all $t \geq 0$ and $x(t) \to 0$ as $t \to +\infty$, and in addition it holds that (Lyapunov stability): for each $\varepsilon > 0$ there exists $\delta > 0$ such that for each $x_0 \in \mathbb{R}^n$ with $|x_0| < \delta$ there is a control u like this and such that $|x(t)| < \varepsilon$ for all $t \geq 0$. We also assume that there are compact subsets $\mathbb{X}_0$ and $\mathbb{U}_0$ of $\mathbb{R}^n$ and $\mathbb{U}$ respectively such that, if the initial state x_0 satisfies also $x_0 \in \mathbb{X}_0$, then the control can be chosen with $u(t) \in \mathbb{U}_0$ for almost all t. (This last property simply rules out the case in which the only way to control to zero is by using controls that must "go to infinity" as the state approaches the origin.)

The main result in [3] is as follows:

Theorem. *The system (1) is asymptotically controllable if and only if it admits an s-stabilizing feedback.* ∎

The "if" implication is trivial, but the proof of the converse is not, and it is based upon: (a) Techniques from control-Lyapunov functions from [15] and [21]; (b) results of nonsmooth analysis, and in particular the systematic use of proximal subgradients, which are substitutes for the gradient for a non-differentiable function, and were originally developed in nonsmooth analysis for the study of optimization problems (see [2]); (c) the analysis of viscosity super-solutions of Hamilton-Jacobi equations in the sense of [10], or more precisely, the "proximal supersolutions" studied in [4]) and the interesting connections between these concepts developed in [4] and the book [22]; and finally, (d) methods developed in the theory of positional differential games in [12] (these techniques were used together with nonsmooth analysis tools in the construction of discontinuous feedback for differential games of pursuit in [5] and games of fixed duration in [6]).

4 Remarks

Although we only defined s-solutions for closed-loop systems (2), in which k is arbitrary but f is locally Lipschitz, one can extend the notion to deal with *general interconnections of systems operating at discrete and continuous time scales*. For example, if either the controller or the plant incorporate variables whose update occurs at discrete instants $s_0, s_1, \ldots$, and if the sequence of s_i's does not have any finite limit points, then one may define π-trajectories by first obtaining a common refinement of the partition π and the partition given by the s_i's. We omit the straightforward details.

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